**SIMATS ENGINEERING**

**CO3 - ASSESSMENT TOOL - 2: Case Study**

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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| Register Number | **Sanjay kumar soy** |

**COURSE OUTCOME COVERED**

**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**

Question Count: 3

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**Code of Conduct**

I sanjay kumar soy(192311151) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: sanjay kumar soy

**Assessment Tasks - Case Study**

1. **Case Study 1: Smart Retail Inventory Management System**
   A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

   A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

**Case Study Rubric**

| **Criteria** | **Weightage** | **Level 4 - Exemplary** | **Level 3 - Proficient** | **Level 2 - Developing** | **Level 1 - Beginning** |
|---|---|---|---|---|---|
| Understanding of Case | 25% | Clearly identifies all technical and business requirements | Identifies most requirements | Basic understanding only | Poor understanding of case |
| Application of Concepts | 25% | Applies suitable cloud concepts and tools effectively | Good application with minor issues | Partial application | Weak or incorrect application |
| Analysis | 20% | Strong analysis with clear trade-off reasoning | Reasonable analysis | Limited analysis | Minimal analysis |
| Solution Design | 20% | Solution is feasible, practical, and well justified | Reasonable solution with minor gaps | Basic solution | Unclear solution |
| Clarity | 10% | Well-structured and easy to follow | Mostly clear | Somewhat unclear | Poorly organized |

# ANSWER

## Case Study 1: Smart Retail Inventory Management System

A retail chain uses a cloud-based inventory management system to manage products and sales across multiple stores. The stores use POS (Point of Sale) systems and dashboards that communicate with centralized cloud services through Web APIs. Product details, transaction records, and analytics data are stored securely in cloud storage, while platform services are used for demand forecasting and inventory analysis. Security is maintained using VPNs, firewalls, and API gateways.

### Working of the System

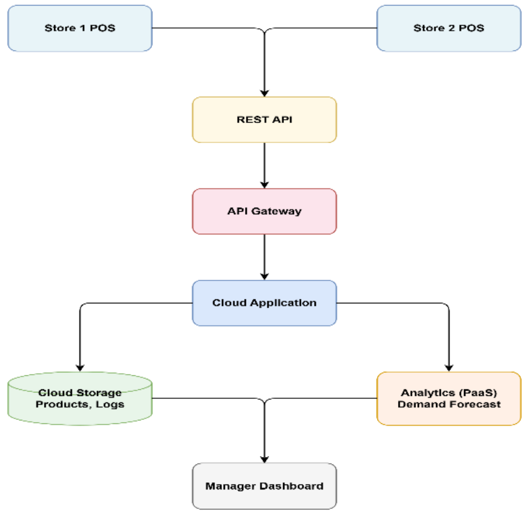
1. Customers purchase products at the store using the POS system.
2. The POS system sends transaction details to the cloud using RESTful Web APIs.
3. The cloud updates the inventory database immediately.
4. Product information, sales records, and transaction logs are stored in cloud storage.
5. Cloud platform services analyze sales data and predict future demand using analytics.
6. Store managers access dashboards to monitor stock levels and sales reports in real time.
7. Security is ensured using VPNs, firewalls, and API gateways, allowing only authorized users and devices to access cloud services.

### Cloud Components Used

- Cloud Storage: Stores product details, transaction logs, and analytics datasets.
- Web APIs: Enable communication between POS systems, dashboards, and cloud services.
- Platform Services (PaaS): Perform analytics, demand forecasting, and reporting.
- Infrastructure (IaaS): Provides virtual servers, networking, and storage resources.
- Security Services: VPN, Firewall, and API Gateway protect data and network communication.

### Advantages of the Solution

- Real-time inventory updates across all stores.
- Prevents stock shortages and overstocking.
- Faster decision-making using cloud analytics.
- Easy to scale by adding more stores without major infrastructure changes.
- Secure communication and data protection.
- Reduced hardware maintenance and operational costs.




**Case Study 2: Cloud-Based Document Collaboration System (10 Marks)**

The enterprise has developed a cloud-based document collaboration system similar to Google Docs. Users access documents through a web browser, where they can edit and share files in real time. Documents are stored in distributed cloud storage with version control, while Web APIs synchronize changes across multiple users and devices. The system also provides high availability, low latency, and strong security using encryption, identity management, and access permissions.

**Working of the System**

1. Users log in to the application through a web browser.
2. The browser sends requests to the cloud using Web APIs.
3. Documents are stored in distributed cloud storage.
4. When one user edits a document, the changes are synchronized instantly to all other users through Web APIs.
5. Version control stores previous versions of documents, allowing users to restore earlier copies if needed.
6. Cloud infrastructure ensures low latency and high availability so users from different locations can collaborate without interruption.
7. Security is maintained using encryption, identity management, and access permissions.

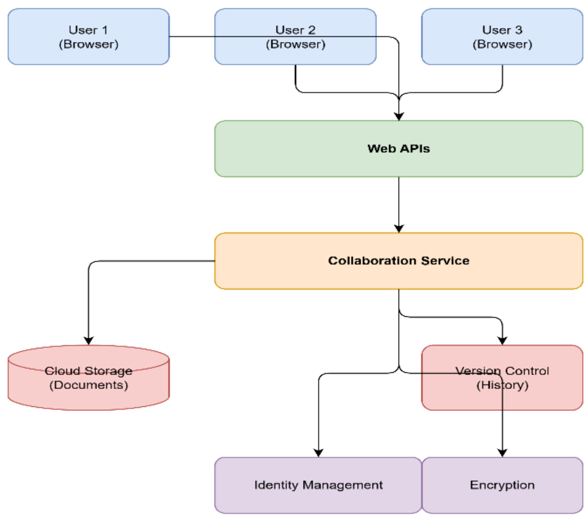
**Cloud Components Used**

- Cloud Storage: Stores documents securely in distributed storage.
- Web APIs: Synchronize document updates between multiple users and devices.
- Platform Services (PaaS): Support collaboration, version control, and synchronization.
- Network Infrastructure: Provides low latency and high availability.

- Security Services: Encryption, Identity Management, and Access Permissions protect user data.

**Advantages of the Solution**

- Enables real-time collaboration among multiple users.
- Automatically synchronizes changes across devices.
- Version control prevents data loss and allows recovery of previous document versions.
- High availability ensures documents can be accessed anytime.
- Encryption and access control protect confidential information.
- Cloud storage allows users to access documents from anywhere with an internet connection.




**Case Study 3: Online Food Ordering Platform**

The startup has developed a cloud-based online food ordering platform that can be accessed through web browsers and mobile applications. Customers interact with the system using RESTful Web APIs, while menu images, invoices, and customer information are stored securely in cloud storage. The platform uses virtual machines and managed Kubernetes for scalability and reliability. Security is provided through authentication tokens, HTTPS, and role-based access control (RBAC).

Working of the System

1. Customers access the food ordering application using a web browser or mobile app.
2. The frontend sends requests to the backend using RESTful Web APIs.
3. The backend processes user requests such as viewing menus, placing orders, and making payments.
4. Cloud storage securely stores menu images, invoices, and customer details.
5. The application runs on Virtual Machines (VMs) and managed Kubernetes, which automatically scale resources based on customer demand.
6. Security is ensured using authentication tokens, HTTPS, and Role-Based Access Control (RBAC) so that only authorized users can access system resources.

**Cloud Components Used**

- Frontend Clients**:** Web browsers and mobile applications.
- RESTful Web APIs**:** Enable communication between clients and backend services.
- Cloud Storage**:** Stores menu images, invoices, and customer data securely.

- Virtual Machines (IaaS): Host backend services.
- Managed Kubernetes**:** Automates deployment, scaling, and management of application containers.
- Security Services: Authentication tokens, HTTPS, and RBAC protect the application.

**Advantages of the Solution**

- Customers can place orders anytime from anywhere.
- Automatic scaling handles high traffic during peak meal hours.
- Cloud storage ensures secure and reliable data storage.
- Kubernetes improves application availability and reliability.
- HTTPS and authentication tokens protect customer information.
- Easy to expand the platform by adding more restaurants and users.

The cloud-based food ordering platform integrates Web APIs, cloud storage, virtual machines, managed Kubernetes, and cloud security services to provide a secure, scalable, and reliable application. This architecture enables real-time order processing, efficient resource management, and an improved customer experience.